

Chao Li, Xiaoming Peng

1. A two-stage fusion strategy and two corresponding modules for comprehensive cross-modal and cross-scale feature fusion.
2. An optimal accuracy-speed trade-off by interleaving Transformer and Mamba layers.
3. SOTA multispectral object detection performance.

The diagram illustrates the architecture of the proposed network, which is designed for visible-light and infrared image fusion. It consists of two main branches: a Visible-light image feature extraction backbone and an Infrared image feature extraction backbone. Both backbones are composed of a series of convolutional layers (C1-C5) and pooling layers (P1-P5). The visible-light backbone starts with a 224x224x3 input, while the infrared backbone starts with a 224x224x1 input. The outputs of the backbones are combined in a Feature fusion module (F1-F5) which uses a 1x1 conv layer. The final output is a Stacked feature map. A legend indicates 'conv' for convolution and 'pool' for pooling. A small inset shows a 'stacked-vis-ali' image.

Figure 1 illustrates the architecture of the proposed channel-wise multi-head cross-attention network. The diagram is divided into six parts: (a) CMT, (b) CLF, (c) Learnable WeightPool, (d) ChannelWiseSample, (e) ChannelDownsample, and (f) Spatial Upsample. A legend indicates that green circles represent dimension-wise addition and blue squares represent channel-wise multiplication.

- (a) CMT:** Shows the Channel-wise Multi-head Cross-Attention module. It takes inputs I^c and I^r and produces outputs I_{out}^c and I_{out}^r . The process involves multi-head cross-attention and a residual connection.
- (b) CLF:** Shows the Channel-wise Local Fusion module. It takes inputs I_1^c, I_2^c, I_3^c and produces outputs I_1^c, I_2^c, I_3^c . The process involves local fusion and a residual connection.
- (c) Learnable WeightPool:** Shows the Learnable WeightPool module. It takes inputs I_1^c, I_2^c, I_3^c and produces outputs I_1^c, I_2^c, I_3^c . The process involves learnable weight pooling and a residual connection.
- (d) ChannelWiseSample:** Shows the ChannelWiseSample module. It takes inputs I_1^c, I_2^c, I_3^c and produces outputs I_1^c, I_2^c, I_3^c . The process involves channel-wise sampling and a residual connection.
- (e) ChannelDownsample:** Shows the ChannelDownsample module. It takes inputs I_1^c, I_2^c, I_3^c and produces outputs I_1^c, I_2^c, I_3^c . The process involves channel-wise downsampling and a residual connection.
- (f) Spatial Upsample:** Shows the Spatial Upsample module. It takes inputs I_1^c, I_2^c, I_3^c and produces outputs I_1^c, I_2^c, I_3^c . The process involves spatial upsampling and a residual connection.

(a) FFL Block

(b) Transition Layer

(c) Mainline Layer

Dataset	Method	mAP50	mAP
FLIR	YOLOv5(SIR)	73.9	39.5
	Fusion-Mamba(YOLOv5)	84.3	44.4
	FTMDet (ours)	84.7 (3.8) 80.4	46.5 (7.0) 72.1
LLVIP	YOLOv5(SIR)	94.6	61.9
	LRAF-Net	97.9	66.6
	FTMDet (ours)	98.0 (3.4) 90.1	68.2 (6.3) 71.8
MFD	YOLOv5(SIR)	77.7	-
	ICAFusion	89.5	68.1
	FTMDet (ours)	90.9 (13.2) 11.4	63.6 (-) 13.5

Highlights

Joint Transformer and Mamba Fusion for Multispectral Object Detection

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- We proposed a staged fusion strategy to achieve more comprehensive feature fusion
- The combination of Transformer and Mamba to reach an optimal accuracy-speed trade-off
- The proposed framework achieves state-of-the-art multispectral detection performance

Joint Transformer and Mamba Fusion for Multispectral Object Detection

Chao Li^a, Xiaoming Peng^{a,*}

^a*School of Automation Engineering, University of Electronic Science and Technology of
China, Chengdu, 611731, SiChuan, China*

Abstract

Multispectral object detection is generally considered better than single-modality-based object detection, due to the complementary properties of multispectral image pairs. However, how to integrate features from images of different modalities for object detection is still an open problem. In this paper, we propose a new multispectral object detection framework based on the Transformer and Mamba architectures, called the joint Transformer and Mamba detection (JTMDet). Specifically, we divide the feature fusion process into two stages, the intra-scale fusion stage and the inter-scale fusion stage, to comprehensively utilize the multi-modal features at different scales. To this end, we designed the so-called cross-modal fusion (CMF) and cross-level fusion (CLF) modules, both of which contain JTM blocks. A JTM block interweaves the Transformer and Mamba layers to robustly capture the useful information in multispectral image pairs while maintaining high inference speed. Extensive experiments on three publicly available datasets conclusively show that the proposed JTMDet framework achieves state-of-the-art multispectral object detection performance, and is competitive with current leading methods.

Keywords: Multispectral object detection, Transformer, Mamba, Feature fusion, Visible-infrared images

*Corresponding author.

Email addresses: 202222060406@std.uestc.edu.cn (Chao Li),
pengxm@uestc.edu.cn (Xiaoming Peng)

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1. Introduction

Object detection has drawn great attention from the computer vision community, with most studies focusing on visible light images rich in textures and colors. However, the quality of visible light images is prone to low brightness when they are captured in rainy and foggy days or at night. Infrared images, on the other hand, characterize the thermal radiation emitted by objects and hence are not affected by harsh illumination conditions. The complementarity between infrared and visible light images makes multispectral image detection a natural demand. Combining information from these two modalities can improve the perceptibility, reliability and robustness of object detection cross various scenarios, as has been proven by recent studies [1–7].

Earlier studies on multispectral objection detection are based on convolutional neural networks (CNNs), mostly adopting a dual-stream architecture [8–10]. Though has achieved a great success in visible image objection detection [11–13], their local receptive field characteristics inherently limits their application in multispectral object detection, which heavily relies on a deep interaction between the two spectral images. For this reason, most recent studies ([1, 2, 14, 15]) opt for the Transformer [16] structure for image feature extraction and interaction, thanks to its cross-space long-range modeling capability. However, the global attention calculations in the Transformer structure also bring heavier computational loads. Mamba [17] is a very recently proposed structure that explains the modeling process from the perspective of state-space equations and uses selective mechanisms and efficient hardware awareness for its implementation. The Mamba structure was originally proposed for the field of natural language processing (NLP) before being introduced into the field of computer vision [18–20]. Unlike the Transformer, the Mamba architecture can maintain linear complexity while being able to handle long distance relationships. However, the Mamba is still outperformed by the Transformer should they have a similar model size [21]. Therefore, how to make the most of the two different structures to create a new structure is still an open problem.

Besides, how to fully interact between different modal features to obtain complementary information from the two types of modalities is also key for multispectral object detection. Most current methods limit the fusion and interaction of multispectral image features to a certain scale [1, 2, 20]. Features of one modality are generally interacted with their same-scale counterparts

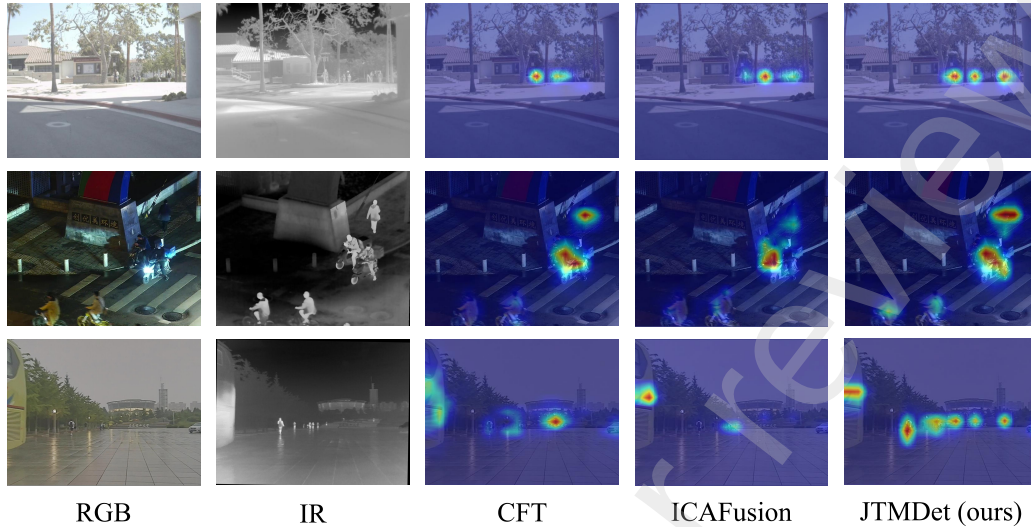


Figure 1: Visualization of P5 heatmaps generated by various cross-modal object detection methods (CFT [1], ICAFusion [2] and the proposed JTMDet) using Layer-GAM [22]. Please note that the P5 heatmaps generated by CFT [1] and ICAFusion [2] failed to activate some obvious object regions. By contrast, the proposed JTMDet can locate the objects more accurately. Zoom in for more details.

in the other modality. However, the impact of object size necessitates a consideration: confining the feature interaction of multispectral images to a fixed scale can result in suboptimal feature interactions. As shown in Fig. 1, some obvious object regions in the P5 heatmaps generated by two previous methods, CFT [1] and ICAFusion [2], are not activated. This is because both CFT and ICAFusion restrict feature interaction solely to the P5 feature map scale, leading to insufficient interaction between large-scale and small-scale object features and hence poor detection of small objects in the scene. In contrast, the proposed method interacts features across multiple scales so as to effectively locate objects of different sizes in the scene.

To address these issues, in this paper we propose a new multispectral object detection framework, named joint Transformer and Mamba fusion for multispectral detection (JTMDet). Specifically, we first come up with a staged multispectral image feature fusion process, which includes an intra-scale feature fusion stage and an inter-scale feature fusion stage. The cross-modal and cross-scale information, which is complementary, is captured separately in these two stages to enhance the quality of the fusion features. To this end, we proposed the so-called cross-modal fusion (CMF) and cross-level

fusion (CLF) modules to implement intra-scale and inter-scale fusion, respectively. Both modules contain JTM blocks, each of which is formed by interweaving the Transformer and Mamba layers. With the powerful modeling capabilities of the Transformer and Mamba layers, the complementary features between different modalities and scales can be better fused. We investigated different combinations of the Transformer and Mamba layers to reach an optimal trade-off between object detection accuracy and speed. Finally, we conducted extensive experiments on three publicly available benchmark datasets to test the proposed approach and compare with other state-of-the-art methods.

In summary, the main contributions of the paper are threefold:

- We proposed a staged feature fusion strategy including the intra-scale fusion stage and the inter-scale fusion stage (Section 3.2). Compared with some other methods that only focus on intra-scale cross-modal fusion ([1], [2]), our two-stage fusion strategy can make full use of features of different scales to further improve the quality and robustness of fusion features.
- We propose two new modules—the so-called CMF and CLF modules—to implement the above-mentioned two-stage feature fusion strategy (Sections 3.3–3.5). Both modules contain JTM blocks, which combine the Transformer and Mamba layers to achieve an optimal accuracy-speed trade-off.
- We conducted extensive experiments and manifest that the proposed JTMDet framework is competitive with current state-of-the-art multispectral object detection approaches (Section 4).

The rest of the paper is organized as follows. In Section 2, we discuss the work most relevant to ours. Then, in Section 3, we describe the proposed JTMDet framework in detail. Next, in Section 4, we compare the proposed approach with state-of-the-art multispectral object detection methods and present ablation study results. Finally, we conclude the paper in Section 5.

2. Related works

In this section, we briefly discuss research studies most related to our work, including multispectral objection detection, the Transformer and Mamba architectures and the hybrid architectures of them.

2.1. Multispectral Object Detection

With the development and popularity of CNNs [11, 23, 24] and the Transformer [25, 26], multispectral object detection has seen great success. Depending on at which level the fusion occurs, current methods can be mainly divided into three categories: pixel-level based, decision-level based, and feature-level based. Pixel-level based methods fuse the input multimodal image pairs into a new fused image that retains the complementary information; the new image is used as input to an object detector. Tang *et al.* addressed the problem of texture information degradation or loss in low-light conditions by incorporating illumination awareness into the fusion process [27]. Sun *et al.* used channel attention to capture key channel features and used adaptive balancing factors to ensure the preservation of instance and texture details [28]. Gao *et al.* used feature-guided hierarchical channel enhancement to avoid information loss during feature extraction [29]. Tian *et al.* proposed an attention-guided transformer module and a contrast enhancement module to promote global information extraction and texture information preservation [30]. Obviously, the focus of pixel-level based approaches is on the quality of the fused image. Decision-level fusion methods apply two separate object detectors on the individual input images, and then fuse the two detection results by some mathematical rules. Li *et al.* proposed a confidence fusion method based on the Dempster-Shafer theory, combining multi-modal features according to the modified confidence [31]. Chen *et al.* proposed a non-learning probabilistic ensembling technique that handles the problem of modality missing [3]. Li *et al.* introduced evidence deep learning (EDL) into multispectral object detection to improve the robustness and reliability of their approach [32]. Several studies have demonstrated that modeling aleatoric uncertainty is beneficial for object detection tasks [33–35]. Compared with the previous two types of approaches, feature-level based approaches are perhaps the mostly researched. This type of methods can be further divided into early-fusion based, mid-fusion based and late-fusion based according to where feature interaction and fusion happens in the network architecture. Wagner *et al.* proved that late fusion is better than early fusion for multispectral object detection [4]. Liu *et al.* further studied four different fusion strategies: early fusion, mid fusion, late fusion and confidence fusion [36]. Their empirical results show that mid fusion works the best, mainly because this strategy retains both rich semantic information and details. Subsequent studies also support their findings and proposed CNN-based feature interaction modules to better utilize the

potential complementarity of multispectral images [5, 9, 37]. Additionally, some approaches introduced the Transformer-based fusion modules to capture more global complementary relationships between multispectral images [1, 2, 14]. Rather than directly fusing image features, some approaches use additional information, such as the scene’s illumination intensity for feature fusion. Zhou *et al.* first used feature differentiation to achieve mutual complementation of features, and then re-weighted the feature maps according to the lighting conditions [10]. Yang *et al.* proposed a bidirectional multi-level fusion strategy that leverages illumination intensity information to progressively enhance the robustness of multi-spectral features against variations in lighting conditions [38].

Considering the fact that the level of feature redundancy varies with images of different modalities, some studies introduced bounding box level semantic segmentation ([5, 9, 37]) or region of interest (ROI) prediction ([8, 39]) to guide the fusion of different regions. Zhang *et al.* proposed a semantic information mask-guided attention fusion module to dynamically weigh and fuse intra-spectral and inter-spectral features [5]. Zhang *et al.* proposed a channel similarity theory that reweights the channels using the differences between the feature prediction mask and the true target box area [40]. To solve the misalignment problem between multimodal images, Zhang *et al.* used a region feature alignment (RFA) module to explicitly align multi-modal features with the aid of additional manually annotated pairs of bounding boxes [8]. Kim *et al.* manually constructed non-completely overlapping image pairs through geometric transformation and used a multi-label learning method to learn to perceive the state of input image pair [41].

2.2. Transformer

Unlike the conventional CNN structure that possesses only a local modeling ability due to its local receptive field, the Transformer [16] has a global modeling ability. Since the pioneering work of ViT [25], several studies [1, 2, 6, 42] have incorporated transformer-based fusion modules into their multispectral object detection architectures. Fang *et al.* [1] directly applied the transformer to a mixed sequence consisting of multispectral image features to capture both intra-modal and inter-modal relationships. Shen *et al.* used a query-guided cross-attention mechanism to directly query useful features and perform cyclic refinement [2]. Chen *et al.* further used the illumination weight as a supervisory factor to determine the proportions of preserved fused features and the original features, which were then fused

by a Transformer structure [14]. Since the Transformer-based cross-modal fusion is computationally intensive with quadratic time and space complexity, although these methods have achieved excellent results in multispectral object detection, they are plagued by the contradiction between detection performance and huge computational resource requirements.

2.3. Mamba

Mamba [17], an improvement on structured state-space sequence model (S4) [43], is a state space model (SSM), it introduces carefully designed hidden states to handle long-distance dependencies while maintaining the parallelism of training and the efficiency of inference. For this reason, it is particularly advantageous to processing long sequential data. Since its introduction in the field of NLP, it has been successfully used to solve various computer vision tasks. Vision Mamba (Vim) introduces a novel general visual backbone, which embeds positions into image sequences and compresses the visual representation via a bidirectional state-space model [18]. The Visual state space model (Vmamba) developed a 2D-selective-scan (SS2D) module to bridge the gap between the ordered nature of 1D signal and the non sequential structure of 2D data [44]. VM-UNet [45] combined Mamba blocks with the Unet [46] framework to enhance medical image segmentation. Video Mamba extended Vmamba to 3D [47]. Some studies have investigated the role of Mamba in the field of multispectral object detection. Fusion-Mamba is perhaps the first study that applied Mamba to multispectral object detection and proposed the Fusion-Mamba block, which reduced the differences between cross-modal features and enhanced the representation consistency of fused features [48]. CFMW [20] combines the Mamba structure with the diffusion model [49], significantly improving the multispectral object detection performance under the influence of adverse weather conditions. Although Mamba-based methods have achieved remarkable results, some studies point out that they still cannot outperform Transformer-based methods of comparable size [21].

2.4. Hybrid architecture of Mamba and Transformer

Some studies have also pointed out that the Transformer and Mamba are complementary structures, and the combination of them can make the most of them [21, 50–53]. Liber *et al.* proposed a new large-scale language model based on a novel hybrid Transformer-Mamba architecture [21]. Fei *et al.* explored a hybrid architecture of the Transformers and Mamba in diffusion models [52]. Wang *et al.* used a hybrid framework of the Transformers and

Mamba for 3D point cloud analysis [53]. Similarly, while existing multispectral object detection methods have shown notable achievements, they are restricted to using either pure Mamba [20, 48] or pure Transformer [1, 2] as the underlying model. How to simultaneously apply Mamba and Transformer to construct a more robust model for addressing the challenges in multispectral object detection remains an open problem and deserves further exploration.

In view of this, our work is pioneering in exploring the combination of Transformer and Mamba blocks for multispectral object detection. By integrating Transformer and Mamba and employing a stage-by-stage fusion strategy, we can achieve better detection accuracy while keeping the computational overhead reasonably low.

3. Proposed JTMDet framework

In this section, we elaborate the proposed JTMDet framework. We first provide some preliminaries on state space models (SSMs) and the Transformer structure in Section 3.1. Then, we give an overview of the JTMDet framework in Section 3.2. Next, we describe in detail the proposed JTM block, the CMF and CLF modules, and the loss function through Sections 3.3–3.5.

3.1. Preliminaries

SSMs. SSM-based models, such as the S4 [43] and the selective state-space model Mamba [17], were inspired by continuous systems and are usually considered to be a linear time-invariant system. This system maps a 1D input $x(t) \in \mathbb{R}$ to an output $y(t) \in \mathbb{R}$ through a hidden state $h(t) \in \mathbb{R}^N$. The system can be mathematically represented by a linear ordinary differential equation (ODE) as follows,

$$\begin{aligned} h'(t) &= \mathbf{A}h(t) + \mathbf{B}x(t) \\ y(t) &= \mathbf{C}h(t) \end{aligned} \tag{1}$$

where $\mathbf{A} \in \mathbb{R}^{N \times N}$ is the evolution matrix and $\mathbf{B} \in \mathbb{R}^{N \times 1}$, $\mathbf{C} \in \mathbb{R}^{N \times 1}$ are two projection matrices.

Since SSMs are derived from continuous systems, they need to be discretized before they can be used in digital system. To this end, the zero-order hold (ZOH) method is used to transform the continuous parameters $\mathbf{A}$ and

231 $\mathbf{B}$ into discrete ones, $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, via the time scale parameter Δ :

$$\begin{aligned}\bar{\mathbf{A}} &= \exp(\Delta \mathbf{A}) \\ \bar{\mathbf{B}} &= (\Delta \mathbf{A})^{-1}(\exp(\Delta \mathbf{A}) - \mathbf{I}) \cdot \Delta \mathbf{B}\end{aligned}\quad (2)$$

232 After obtaining $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$, we can derive the following discrete version of
233 Eq. (1):

$$\begin{aligned}h_t &= \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t \\ y_t &= \mathbf{C}h_t\end{aligned}\quad (3)$$

234 At last, the output of an SSM model can be calculated using a global
235 convolution, as shown in Eq. (4):

$$\begin{aligned}\bar{\mathbf{K}} &= (\mathbf{C}\bar{\mathbf{B}}, \mathbf{C}\bar{\mathbf{A}}^{-1}\bar{\mathbf{B}}, \dots, \mathbf{C}\bar{\mathbf{A}}^{(M-1)}\bar{\mathbf{B}}) \\ \bar{\mathbf{A}}^{(M-1)} &= \underbrace{\bar{\mathbf{A}} \times \bar{\mathbf{A}} \cdots \times \bar{\mathbf{A}}}_{M-1} \\ \mathbf{y} &= \mathbf{x} * \bar{\mathbf{K}}\end{aligned}\quad (4)$$

236 where M is the length of the input sequence $\mathbf{x}$, and $\bar{\mathbf{K}} \in \mathbb{R}^M$ represents a
237 structured convolutional kernel.

238 **The Transformer.** The Transformer structure was originally designed
239 for NLP [16]. It performs global self-attention calculations on an input 1D
240 sequence, $\mathbf{x} \in \mathbb{R}^{L \times C}$, to aggregate global information of the input through
241 attention weights. The whole process can be expressed as follows: The input
242 $\mathbf{x}$ is first projected by three weight matrices, $\mathbf{W}^Q$, $\mathbf{W}^K$ and $\mathbf{W}^V$, to compute
243 a set of queries($\mathbf{Q}$), keys($\mathbf{K}$), and values($\mathbf{V}$), as follow:

$$\begin{aligned}\mathbf{Q} &= \mathbf{x}\mathbf{W}^Q \\ \mathbf{K} &= \mathbf{x}\mathbf{W}^K \\ \mathbf{V} &= \mathbf{x}\mathbf{W}^V\end{aligned}\quad (5)$$

244 where $\mathbf{W}^Q \in \mathbb{R}^{C \times d_q}$, $\mathbf{W}^K \in \mathbb{R}^{C \times d_k}$ and $\mathbf{W}^V \in \mathbb{R}^{C \times d_v}$ are learnable weight
245 matrices, and d_q , d_k , d_v dictate the dimensions of the three matrices, $\mathbf{Q}$ and
246 $\mathbf{K}$ and $\mathbf{V}$. Generally, they take equal values, *i.e.* $d_q = d_k = d_v = d_m$. Then,
247 the output matrix of attention, $\mathbf{Z}$, is computed as:

$$\mathbf{Z} = \text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_m}}\right)\mathbf{V}\quad (6)$$

where $\sqrt{d_m}$ is a scaling factor used to prevent the softmax function from being trapped into a region with a small gradient as the magnitude of dot products grow larger.

Usually, a multi-head attention mechanism is adopted, which linearly projects the queries, keys and values h times with different learnable linear projections. The multi-head attention mechanism is formulated as:

$$\begin{aligned}\mathbf{Z}' &= \text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) \\ &= \text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_h) \mathbf{W}^Z \\ \mathbf{Z}_i &= \text{Attention}(\mathbf{QW}_i^Q, \mathbf{KW}_i^K, \mathbf{VW}_i^V)\end{aligned}\quad (7)$$

where the $\mathbf{Z}_h \in \mathbb{R}^{L \times d_h}$ is the output matrix of the i -th attention head, $\text{Concat}(\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_h) \in \mathbb{R}^{L \times hd_h}$ represents the concatenation of $\mathbf{Z}_1, \mathbf{Z}_2, \dots, \mathbf{Z}_h$. The projections matrices $\mathbf{W}_i^Q, \mathbf{W}_i^K$, and $\mathbf{W}_i^V$ are of dimension $\mathbb{R}^{d_m \times d_h}$ and $\mathbf{W}^Z \in \mathbb{R}^{hd_h \times d_m}$, and $d_h = d_m/h$ is the dimension of each attention head.

Then the Transformer uses a multilayer perceptron (MLP) with a Gaussian Error Linear Units (GELU)[54] activation in between to calculate the output sequences $\mathbf{y} \in \mathbb{R}^{L \times C}$, which has the same shape as the input sequence $\mathbf{x}$:

$$\mathbf{y} = \text{MLP}(\mathbf{Z}'') + \mathbf{Z}'' = \text{FC}_1(\text{GELU}(\text{FC}_2(\mathbf{Z}''))) \quad (8)$$

where $\mathbf{Z}'' = \mathbf{Z}' + \mathbf{x}$, and $\text{MLP}(\bullet)$ consists of two fully connected feed-forward networks, FC_1 and FC_2 .

3.2. Overview

The proposed JTMDet consists of two stages: intra-scale fusion stage and inter-scale fusion stage. Its overall architecture is shown in Fig. 2. In the intra-scale fusion stage, the multi-modal features of the two images of different modalities are interacted through the CMF module to hopefully to mutually enhance each other and themselves. In the inter-scale fusion stage, the framework simultaneously integrates feature maps of different scales by the CLF module and enhances them again. For a fair comparison with other multispectral object detection methods, we use the same backbone network of YOLOv5 [55].

Consider an visible-infrared image pair $\{\mathbf{I}^V, \mathbf{I}^R\}$, the object detection process can be described as follows:

First, the multispectral images are input into the two branches, $\mathbf{h}_i$ and φ_i ($i=1-5$), of the dual-stream feature extraction backbone, respectively. We

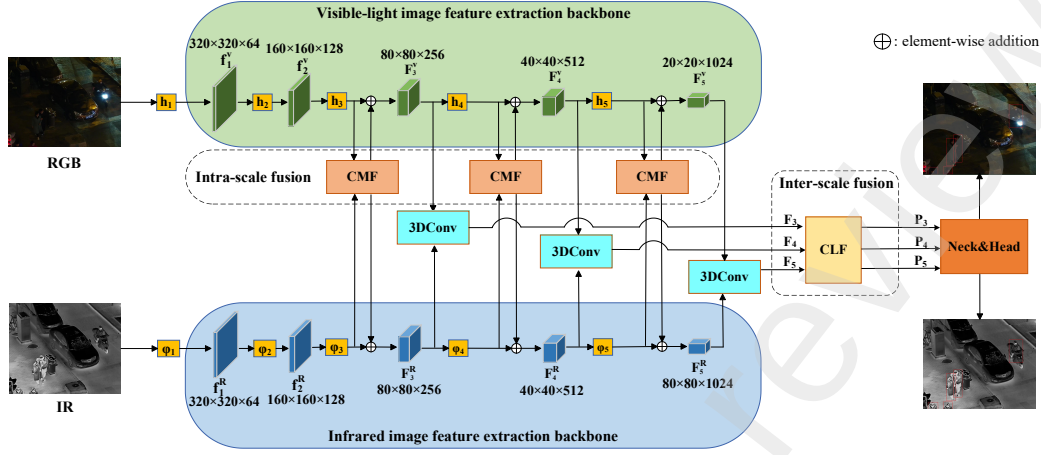


Figure 2: The overall architecture of the proposed JTMDet framework. This framework consists of a dual-stream feature extraction network, three CMF modules, three fusion modules, and a CLF module. The neck and head were borrowed from YOLOv5 [55]. The image pairs are first processed by the dual-stream backbone network, $\mathbf{h}_i$ and φ_i ($i=1-5$), to obtain feature maps $\mathbf{f}_i^V$ and $\mathbf{f}_i^R$. The feature maps of the last three levels are fed into CMF in pairs to obtain the enhanced feature maps $\mathbf{F}_i^V$ and $\mathbf{F}_i^R$ ($i=3-5$). After that, a 3D convolution module is used to generate the fused feature maps $\mathbf{F}_i$ ($i=3-5$) from the enhanced feature maps. Then, the fused feature maps of different scales are sent to the CLF module together for further enhancement to obtain the input $\mathbf{P}_i$ ($i=3-5$) of neck and head network.

use $\mathbf{f}_i^V$ and $\mathbf{f}_i^R$ to denote feature maps of the i -th layer of $\mathbf{h}$ and φ , as follow:

$$\begin{aligned}\mathbf{f}_i^V &= \mathbf{h}_i(\mathbf{f}_{i-1}^V) \\ \mathbf{f}_i^R &= \varphi_i(\mathbf{f}_{i-1}^R)\end{aligned}\tag{9}$$

In the intra-scale stage, the feature maps of the last three layers ($i=3-5$) are sent into the cross-modal fusion module CMF to integrate complementary information from each other. Then, the complementary features $\bar{\mathbf{f}}_i^V$ and $\bar{\mathbf{f}}_i^R$ are added to their original feature maps $\mathbf{f}_i^V$ and $\mathbf{f}_i^R$, respectively, to obtain the intra-scale enhanced feature maps, $\mathbf{F}_i^V$ and $\mathbf{F}_i^R$, as follow:

$$\begin{aligned}\bar{\mathbf{f}}_i^V, \bar{\mathbf{f}}_i^R &= \text{CMF}(\mathbf{f}_i^V, \mathbf{f}_i^R) \\ \mathbf{F}_i^V &= \mathbf{f}_i^V + \bar{\mathbf{f}}_i^V \\ \mathbf{F}_i^R &= \mathbf{f}_i^R + \bar{\mathbf{f}}_i^R\end{aligned}\tag{10}$$

Then, we generate a fused feature maps $\mathbf{F}_i$ ($i=3-5$) from the intra-scale

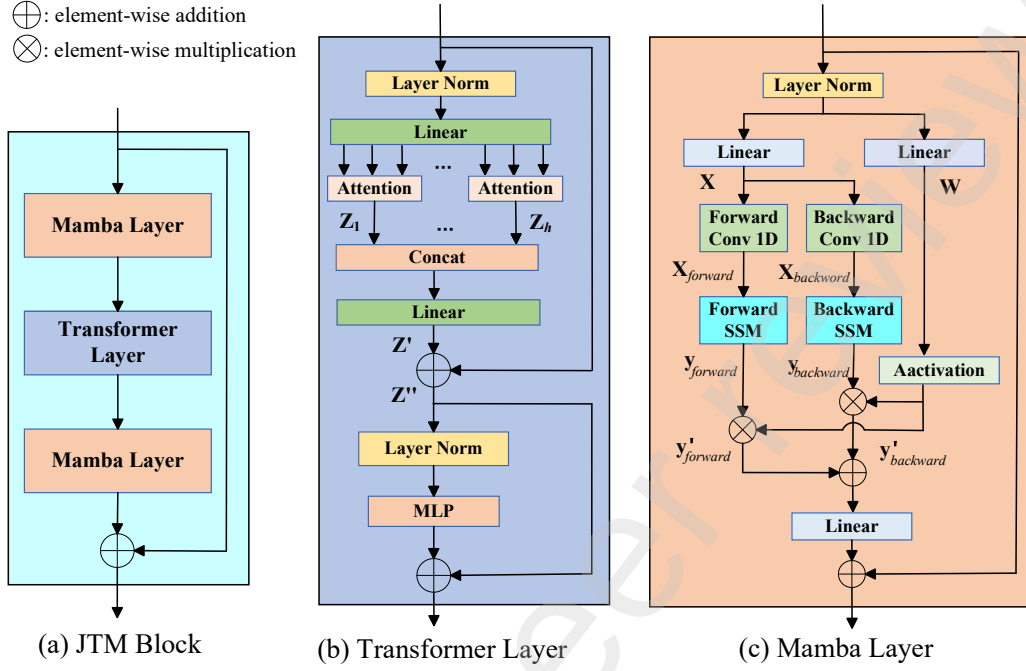


Figure 3: The structure of JTMBlock and the structures of the Transformer [25] layer and Mamba [18] layer used in it.

enhanced feature maps $\mathbf{F}_i^V$ and $\mathbf{F}_i^R$ through a 3D convolution operation

$$\mathbf{F}_i = 3DConv(\mathbf{F}_i^V, \mathbf{F}_i^R) \quad (11)$$

In the inter-scale fusion stage, we input the $\mathbf{F}_i$ feature maps into the CLF module, in which these features of different scales are enhanced, to finally obtain the inter-scale enhanced feature maps $\mathbf{P}_i$ used for the detection network, as follows:

$$\mathbf{P}_3, \mathbf{P}_4, \mathbf{P}_5 = CLF(\mathbf{F}_3, \mathbf{F}_4, \mathbf{F}_5) \quad (12)$$

Now we detail the constituents of the proposed JTMDet framework in the subsequent subsections.

3.3. JTM Block

Each JTM block, shown in Fig. 3 (a), consists two Mamba layers and one Transformer layer. The Transformer and Mamba layers, shown in Fig.

3 (b) and Fig. 3 (c), are borrowed from ViT [25] and Vim [18], respectively. Since the Transformer structure has been delineated in Section 3.1, here we will focus on the Mamba structure shown in Fig. 3 (c). Specifically, for an input sequence $\mathbf{x}$, it is first normalized by the normalization layer before being linearly projected into $\mathbf{X}$ and $\mathbf{W}$ using two linear projections, $\text{Linear}^{\mathbf{X}}$ and $\text{Linear}^{\mathbf{W}}$.

$$\begin{aligned}\mathbf{X} &= \text{Linear}^{\mathbf{X}}(\text{Norm}(\mathbf{x})) \\ \mathbf{W} &= \text{Linear}^{\mathbf{W}}(\text{Norm}(\mathbf{x}))\end{aligned}\quad (13)$$

Then, $\mathbf{X}$ is processed separately from the forward and backward directions. For direction $\mathbf{o} = \{\text{forward}, \text{backward}\}$, a one-dimensional convolution with Sigmoid-weight Linear Unit (SiLU)[56] is first applied to $\mathbf{X}$ to get $\mathbf{X}_{\mathbf{o}}$. Then, $\mathbf{X}_{\mathbf{o}}$ is linearly projected to $\mathbf{B}_{\mathbf{o}}$, $\mathbf{C}_{\mathbf{o}}$ and $\Delta_{\mathbf{o}}$ by $\text{Linear}_{\mathbf{o}}^{\mathbf{B}}$, $\text{Linear}_{\mathbf{o}}^{\mathbf{C}}$ and $\text{Linear}_{\mathbf{o}}^{\Delta}$, respectively, as follow:

$$\begin{aligned}\mathbf{X}_{\mathbf{o}} &= \text{SiLU}(\text{Conv1d}(\mathbf{X})) \\ \mathbf{B}_{\mathbf{o}} &= \text{Linear}_{\mathbf{o}}^{\mathbf{B}}(\mathbf{X}_{\mathbf{o}}) \\ \mathbf{C}_{\mathbf{o}} &= \text{Linear}_{\mathbf{o}}^{\mathbf{C}}(\mathbf{X}_{\mathbf{o}}) \\ \Delta_{\mathbf{o}} &= \log(1 + \exp(\text{Linear}_{\mathbf{o}}^{\Delta}(\mathbf{X}_{\mathbf{o}}) + \text{Parameter}_{\mathbf{o}}^{\Delta}))\end{aligned}\quad (14)$$

where $\text{Parameter}_{\mathbf{o}}^{\Delta}$ is a learnable paramter.

After that, the timescale parameter $\Delta_{\mathbf{o}}$ is used to transform the $\overline{\mathbf{A}}_{\mathbf{o}}$ and $\overline{\mathbf{B}}_{\mathbf{o}}$, which are the directed versions of $\mathbf{A}_{\mathbf{o}}$ and $\mathbf{B}_{\mathbf{o}}$, respectively.

$$\begin{aligned}\overline{\mathbf{A}}_{\mathbf{o}} &= \Delta_{\mathbf{o}} \oplus \text{Parameter}_{\mathbf{o}}^{\mathbf{A}} \\ \overline{\mathbf{B}}_{\mathbf{o}} &= \Delta_{\mathbf{o}} \oplus \mathbf{B}_{\mathbf{o}}\end{aligned}\quad (15)$$

where $\text{Parameter}_{\mathbf{o}}^{\mathbf{A}}$ is a learnable paramter.

The output $\mathbf{y}_{\mathbf{o}}$ is then computed through the $\text{SSM}(\bullet)$, which corresponds to Eq. (2) and Eq. (3) in Section 3.1.

$$\mathbf{y}_{\mathbf{o}} = \text{SSM}(\overline{\mathbf{A}}_{\mathbf{o}}, \overline{\mathbf{B}}_{\mathbf{o}}, \mathbf{C}_{\mathbf{o}})\mathbf{X}_{\mathbf{o}}\quad (16)$$

By applying Eqs. (14–16) to the forward and backward processes, *i.e.* $\mathbf{o} = \text{forward}$ and $\mathbf{o} = \text{backward}$, we can obtain the forward output $\mathbf{y}_{\text{forward}}$ and the backward output $\mathbf{y}_{\text{backward}}$. Finally, they are gated by $\mathbf{W}$ and added together. The output token sequence, $\mathbf{y}$, is obtained using Eq. (17):

$$\begin{aligned}\mathbf{y}'_{\text{forward}} &= \mathbf{y}_{\text{forward}} \otimes \text{SiLU}(\mathbf{W}) \\ \mathbf{y}'_{\text{backward}} &= \mathbf{y}_{\text{backward}} \otimes \text{SiLU}(\mathbf{W}) \\ \mathbf{y} &= \text{Linear}^{\mathbf{y}}(\mathbf{y}'_{\text{forward}} + \mathbf{y}'_{\text{backward}}) + \mathbf{x}\end{aligned}\quad (17)$$

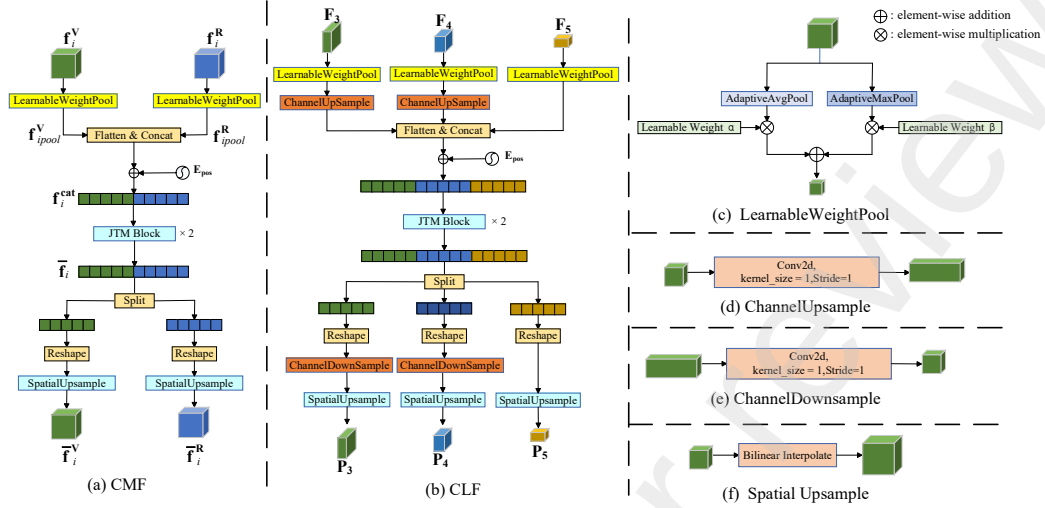


Figure 4: The structure of the cross modality fusion (CMF) module and the cross level fusion (CLF) module.

where Linear^y is a linear projection and $\otimes$ represents element-level multiplication.

3.4. CMF Module

The structure of the CMF module is shown in Fig. 4 (a). Inspired by CFT [1], we first use adaptive pooling layers to downsample the feature maps, $\mathbf{f}_i^V$ and $\mathbf{f}_i^R$, to a fixed resolution of 8×8 to reduce the amount of calculation. However, unlike CFT, which only uses average pooling layers, we adaptively combine average pooling layers $\text{AvgPooling}(\bullet)$ and maximum pooling layers $\text{MaxPooling}(\bullet)$ for feature pooling as shown in Fig. 4 (c):

$$\begin{aligned} \mathbf{f}_{ipool}^V &= \alpha_i^V \text{AvgPooling}(\mathbf{f}_i^V) + \beta_i^V \text{MaxPooling}(\mathbf{f}_i^V) \\ \mathbf{f}_{ipool}^R &= \alpha_i^R \text{AvgPooling}(\mathbf{f}_i^R) + \beta_i^R \text{MaxPooling}(\mathbf{f}_i^R) \end{aligned} \quad (18)$$

where $\alpha_i^V, \alpha_i^R, \beta_i^V$ and β_i^R are learnable weights with value between 0 and 1.

Since both the Mamba and the Transformer layers can only process sequence input, we then flatten the feature maps of the two modalities into sequences and concatenate them into a new sequence. Further, a learnable position embedding $\mathbf{E}_{pos}$ is added to retain the spatial relationship between the original patches of an image, as follow:

$$\mathbf{f}_i^{\text{cat}} = \text{Concat}[\text{flatten}(\mathbf{f}_{ipool}^V), \text{flatten}(\mathbf{f}_{ipool}^R)] + \mathbf{E}_{pos} \quad (19)$$

Then, as Fig. 4 (a) shows, $\mathbf{f}_i^{\text{cat}}$ passes through a sequence of JTM blocks. The resultant feature sequence, $\bar{\mathbf{f}}_i$, is again split into two subsequences, which are further reshaped into two feature maps with resolution 8×8 . Finally, they are upsampled via bilinear interpolation to have the same resolution as the original image (Fig. 4 (f)). The above-mentioned process can be summarized as follows:

$$\begin{aligned}\bar{\mathbf{f}}_i &= \text{JTMBlocks}(\mathbf{f}_i^{\text{cat}}) \\ \bar{\mathbf{f}}_i^{\text{V}}, \bar{\mathbf{f}}_i^{\text{R}} &= \text{Upsample}(\text{Reshape}(\text{Split}(\bar{\mathbf{f}}_i)))\end{aligned}\quad (20)$$

Just like VIT and Vim, the proposed JTMDet can be scaled by directly stacking JTM block. We will investigate the influence of the number of JTM blocks in Section 4.6.

3.5. CLF module

We designed the CLF module to integrate features of various scales. As shown in Fig. 4 (b), the feature maps of different scales are first downsampled to have the same spatial size (8×8 in this paper) using the adaptive combined pooling strategy described in Section 3.4. Then, they are upsampled to have the same number of channels through a 1×1 convolution layer as Fig. 4 (d). Note that $\mathbf{F}_5$ has the largest number of channels, so we align the number of channels of $\mathbf{F}_3$ and $\mathbf{F}_4$ to that of $\mathbf{F}_5$ through a 1×1 convolution layer. Then $\mathbf{F}_3$, $\mathbf{F}_4$, and $\mathbf{F}_5$ are simultaneously sent to JTM blocks for cross-scale information interaction. Finally, as with the CMF module, we restore the processed feature map to its original size through 1×1 convolution and bilinear interpolation as shown in Fig. 4 (e) and Fig. 4 (f).

3.6. Loss Function

The loss function of the proposed JTMDet is consistent with that a YOLO model. The total loss is a combination of three terms, namely, the bounding-box regression loss ($\mathcal{L}_{\text{box}}$), the classification loss ($\mathcal{L}_{\text{cls}}$), and the confidence loss ($\mathcal{L}_{\text{obj}}$),

$$\mathcal{L}_{\text{total}} = \lambda_1 \mathcal{L}_{\text{box}} + \lambda_2 \mathcal{L}_{\text{cls}} + \lambda_3 \mathcal{L}_{\text{obj}} \quad (21)$$

where λ_1 , λ_2 and λ_3 are the weights of the three losses, respectively. Here, $\mathcal{L}_{\text{box}}$ is the generalized intersection over union (GIoU) loss [57]. For $\mathcal{L}_{\text{cls}}$ and $\mathcal{L}_{\text{obj}}$, the cross-entropy and the squared-error losses are adopted, respectively.

360 4. Experiment

361 In this section, we evaluate the performance of the proposed method.
362 First, we introduce the datasets used, the details of the experiments, and
363 the evaluation metrics in Sections 4.1–4.3. Then, we compare the proposed
364 method with existing SOTA methods and demonstrate the effectiveness of
365 the proposed method from both quantitative (Section 4.4) and qualitative
366 (Section 4.5) perspectives. Finally, in Section 4.6, we conduct ablation ex-
367 periments to verify the generalization performance of our proposed method
368 and the role of each component.

369 4.1. Dataset

370 The proposed JTMDet framework was evaluated on three benchmark
371 datasets—FLIR [58], LLVIP [59] and M³FD [60], and compared with some
372 other state-of-the-art approaches for multispectral object detection. A brief
373 description of these datasets is given below:

374 **FLIR.** The FLIR ADAS dataset is a challenging infrared-visible object
375 detection dataset containing objects in day and night. In this study, we
376 experimented with its "aligned" version [9], which contains 5142 pairs of
377 infrared-visible images, of which 4129 image pairs were used for training and
378 1013 pairs for testing. The image resolution is 640×512 . There are three
379 object categories in this dataset: people, cars, and bicycles.

380 **LLVIP.** LLVIP dataset is a visible-infrared paired pedestrian dataset for
381 low-light vision. The dataset contains 15,488 pairs of well-aligned images, of
382 which 12,025 pairs are used for training and 3,463 pairs are used for testing,
383 with a resolution of 1280×1024 . There is only one category in the dataset:
384 person.

385 **M³FD.** This dataset contains 4,200 visible-infrared aligned image pairs
386 with a resolution of 1024×768 . It has six categories: people, car, bus,
387 motorcycle, lamp, and truck. Following [61], we divided the data into a
388 training set of 3360 pairs and a validation set of 840 image pairs.

389 4.2. Metrics

390 We use the two most commonly used evaluation metrics, mAP50 and
391 mAP, to assess the performance of the proposed method. Based on the
392 area detected by the model and the actual ground truth (GT) area, the
393 intersection over union (IoU) of each prediction is calculated by the ratio of
394 the intersection between the area detected by the model and the GT area

to their union. The average precision (AP) of a category can be determined based on the ratio of positive samples to the sum of positive samples and negative samples in the prediction results of this category. When the IOU of a prediction result is greater than the threshold, the result is considered to be a positive sample, otherwise it is considered to be a negative sample. The mAP50 metric represents the average of all AP values of all categories under an IoU threshold of 0.50. The mAP metric measures the average AP under IoU ranges from 0.50 to 0.95 with a step size of 0.05 [7]. Higher values of mAP50 and mAP indicate better model performance.

4.3. Implementation details

All the experiments reported in this paper were conducted using the PyTorch 2.1.1 framework with a sing Nvidia RTX 3090 24G GPU. The YOLOv5 pipeline used in our implementation is originated from [55]. We used a stochastic gradient descent (SGD) optimizer with an initial learning rate of 0.01, a momentum of 0.937, and a weight decay of 0.0005. We reshaped the input image size to 640×640 , and used a batch size of 16. We trained 100 epochs for the FLIR and LLVIP datasets and 200 epochs for the M³FD dataset. For data augmentation, we use the mosaic method [62], which mixes four training images into one image. We used the YOLOv5 model pre-trained on the COCO dataset [63] as the weight initialization of the dual-stream backbone network.

4.4. Quantitative analysis

Comparisons on FLIR. We compared the proposed method with other state-of-the-art multispectral object detection approaches ([1, 2, 5–7, 42, 64]) and two single-modality object detection methods (FasterRCNN [23] and YOLOv5 [55]) on the FLIR dataset. The results are reported in Table 1. Of the multispectral object detection approaches, GAFF [5] guides the model to learn an attention module that fuses features of different modalities through a split branch. Recent work has focused on using Transformer to solve the problem of multispectral target detection. CFT [1] first introduced Transformer into the field of multispectral object detection. CrossFormer [64] and ICAFusion [2] focus on using cross attention for better cross-modal enhancement. MFPT [42] and RSDet [7] combine Transformer with other mechanisms, such as feature pyramids or frequency domain methods. LRAF-Net [6] combines convolution and window attention to improve detection performance by fusing enhanced long-term dependencies of visible and infrared

Table 1: Quantitative analysis of different approaches on the FLIR dataset. **Red**:The best results. **Green**: The second-best results

Methods	Backbone	Data Type	mAP50(%) $\uparrow$	mAP(%) $\uparrow$	Param.(M)	Time(ms)
FasterRCNN[23] (NIPS'2015)	ResNet50	RGB	64.9	28.9	41.13	-
FasterRCNN[23] (NIPS'2015)	ResNet50	IR	74.4	37.6	41.13	-
YOLOv5[55] (2020)	CSPDarknet53	RGB	67.8	37.8	47.06	-
YOLOv5[55] (2020)	CSPDarknet53	IR	73.9	39.5	47.06	-
GAFF[5] (WACV'2021)	ResNet18	RGB+IR	72.9	36.6	-	9.34
CFT[1] (2021)	CSPDarknet53	RGB+IR	78.7	40.2	206	44.4
MFPT[42] (TITS'2023)	ResNet50	RGB+IR	80.0	-	-	50
CrossFormer[64] (PRL'2024)	CSPDarknet53	RGB+IR	79.3	42.1	340	80
RSDet[7] (2024)	CSPDarknet53	RGB+IR	81.1	41.4	-	-
LRAF-Net[6] (TNNLS'2024)	CSPDarknet53	RGB+IR	80.5	42.8	18.8	21.4
ICAFusion[2] (PR'2024)	CSPDarknet53	RGB+IR	79.2	41.1	120.2	30.4
Fusion-Mamba[48] (2024)	CSPDarknet53	RGB+IR	84.3	44.4	244.6	61
Fusion-Mamba[48] (2024)	YOLOv8	RGB+IR	84.9	47.0	287.6	78
JTMDet (ours)	CSPDarknet53	RGB+IR	84.7	46.5	237.4	29.7

features. The Fusion-Mamba [48] introduces the Mamba structure into the field of multispectral detection for the first time, reducing modal differences through a carefully designed structure to achieve better fusion.

It can be seen from Table 1 that for the same object detector, the results are unanimously better obtained on the infrared images than on the visible-light images. This is because the infrared images are less affected by the adverse effects of insufficient illumination. On the other hand, all most all methods (except GAFF) that use both modalities for object detection achieved better performance than single-modality-based methods.

Among the multi-modal-based detection methods used for comparison, GAFF performed the worst, with lower accuracy than YOLOv5 using infrared images alone for object detection (72.9% vs. 73.9%). This is because GAFF uses ResNet18 as the backbone network, which limits its ability to extract features. This result also shows that the quality of features extracted by the backbone network has an important impact on the performance of multispectral detection methods. Among all the methods, the proposed method is second only to Fusion-Mamba based on YOLOv8 [65] in accuracy. We conjecture that this performance gap may come from the imperfectly aligned nature of the FLIR dataset, and the proposed method does not explicitly handle the alignment between the image pairs. As be shown shortly in the experiments on the LLVIP dataset, this performance gap is closed when the visible-infrared pairs are better aligned. Specifically, when both using the same YOLOv5 backbone network, the proposed method achieved a 0.4%

Table 2: Quantitative analysis of different approaches on the LLVIP dataset. **Red**: The best results. **Green**: The second-best results

Methods	Backbone	Data Type	mAP50(%) $\uparrow$	mAP(%) $\uparrow$	Param.(M)	Time(ms)
FasterRCNN[23] (NIPS'2015)	ResNet50	RGB	91.4	49.2	41.13	-
FasterRCNN[23] (NIPS'2015)	ResNet50	IR	96.1	61.1	41.13	-
YOLOv5[55] (2020)	CSPDarknet53	RGB	90.8	50.0	47.06	-
YOLOv5[55] (2020)	CSPDarknet53	IR	94.6	61.9	47.06	-
GAFF[5] (WACV'2021)	ResNet18	RGB+IR	94.0	55.8	-	-
CFT[1] (2021)	CSPDarknet53	RGB+IR	97.5	63.6	206	43.4
ProbEn[3] (ECCV'2022)	ResNet50	RGB+IR	93.4	51.5	-	-
MS-DETR[67] (2023)	ResNet50	RGB+IR	97.9	66.1	-	-
CSSA[66] (CVPR'2023)	ResNet50	RGB+IR	94.3	59.2	-	-
LRAF-Net[6] (TNNLS'2024)	CSPDarknet53	RGB+IR	97.9	66.3	18.8	21.4
ICAFusion[2] (PR'2024)	CSPDarknet53	RGB+IR	97.6	63.6	120.2	30.4
Fusion-Mamba[48] (2024)	CSPDarknet53	RGB+IR	96.8	62.8	244.6	61
Fusion-Mamba[48] (2024)	YOLOv8	RGB+IR	97.0	64.3	287.6	78
JTMDet (ours)	CSPDarknet53	RGB+IR	98.0	68.2	237.4	29.7

mAP50 score and a 2.1% mAP score more than Fusion-Mamba. Meanwhile, in terms of running time, JTMDet has fewer parameters and runs faster than Fusion-Mamba. Compared with CFT, which has a similar structure as ours, JTMDet runs 29% faster and has 15% fewer parameters. JTMDet is as fast as ICAFusion, even if it has about twice the number of parameters as the former.

Comparisons on LLVIP. The results of nine multispectral object detection methods and two single-modal object detection methods on LLVIP are summarized in Table 2. In addition to the methods mentioned above, ProbEn [3] uses a decision-level fusion method to fuse the detection results of two single-modality data. CSSA [66] uses channel switching and spatial attention to effectively fuse inputs from different modality. MS-DETR [67] extends DETR [26] to the field of multimodal detection and achieves good detection results. As ICAFusion was not reported on the LLVIP dataset, we retrained and tested it on the LLVIP dataset using the model provided by the authors.

As can be seen from Table 2, ProbEn had the worst performance among all multispectral detection methods. This is because ProbEn is a decision-level fusion method for probabilistic integration, and its performance depends on the performance of the single-modal method used for integration. MS-DETR and LRAF-NET were runner-ups behind the proposed JTMDet. When the multispectral image pairs are better aligned, as in the case of the LLVIP dataset, JTMDet outperformed Fusion-Mamba based on YOLOv8,

Table 3: Quantitative analysis of different approaches on the M³FD dataset. **Red**: The best results. **Green**: The second-best results

Methods	Backbone	Data Type	mAP50(%) $\uparrow$	mAP(%) $\uparrow$
YOLOv5[55] (NIPS’2015)	CSPDarknet53	RGB	74.6	-
YOLOv5[55] (NIPS’2015)	CSPDarknet53	IR	77.7	-
YOLOv8[65] (2023)	YOLOv8	RGB	80.9	52.5
YOLOv8[65] (2023)	YOLOv8	IR	79.5	53.1
CFT[1] (2021)	CSPDarknet53	RGB+IR	88.6	59.2
RFNet[68] (CVPR’2022)	CSPDarknet53	RGB+IR	79.4	53.2
TarDAL[60] (CVPR’2022)	CSPDarknet53	RGB+IR	80.5	54.1
CDDFuse[69] (CVPR’2023)	CSPDarknet53	RGB+IR	81.1	54.3
IGNet[70] (MM’2023)	CSPDarknet53	RGB+IR	81.5	54.5
ICAFusion[2] (PR’2024)	CSPDarknet53	RGB+IR	89.5	60.1
Fusion-Mamba[48] (2024)	CSPDarknet53	RGB+IR	85.0	57.5
Fusion-Mamba[48] (2024)	YOLOv8	RGB+IR	88.0	61.9
JTMDet (ours)	CSPDarknet53	RGB+IR	90.9	63.6

which had the best performance on the FLIR dataset. When it comes to the mAP score, the proposed JTMDet outperformed the previous best method (LRAF-Net) by $68.2\% - 66.3\% = 1.9\%$.

Comparisons on M³FD. Unlike FLIR and LLVIP, M³FD has been widely used in the field of image fusion, so many methods designed for this dataset are pixel-based methods. Of the methods chosen for comparison, RFNet [68] proposed an unsupervised method to simultaneously process image fusion and registration tasks. TarDAL [60] expands the two-layer optimization formula of the joint problem of fusion and detection into a target-aware dual adversarial learning network. CDDFusion [69] combines CNN and Transformer to form a dual-branch structure, and promotes the extraction of shared features and specific features by driving decomposition loss. IGNet [70] specifically introduces an interactive graph neural network structure to prevent information loss in the fusion process. As CFT and ICAFusion were not reported on the M³FD dataset, we retrained and tested them on the M³FD dataset using the models provided by the respective authors.

As shown in Table 3, the proposed method again achieved the state-of-the-art performance on the M³FD dataset. Among all the multispectral methods, RFNet, TarDaAL, CDDFusion and IGNet focus on the quality of the generated fused images. Though their object detection performance is not outstanding, they still outperformed the corresponding single-modal

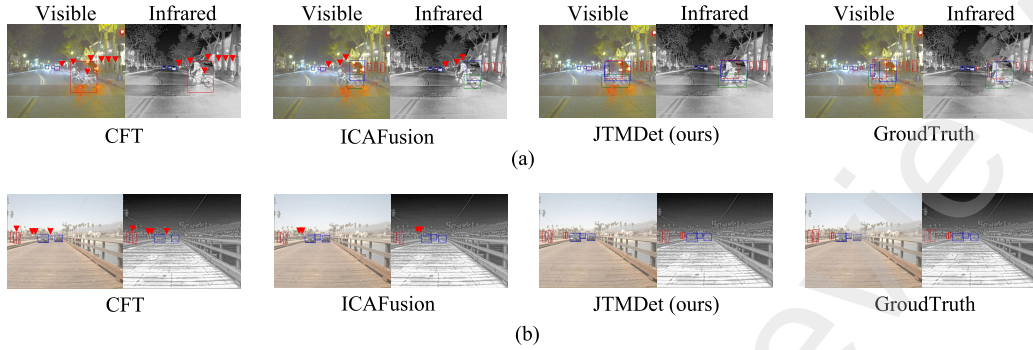


Figure 5: Qualitative analysis of CFT[1], ICAFusion[2] and the proposed method on FLIR dataset. For simplicity, the categories of the predicted results are not shown, and different categories are marked with boxes of different colors. **Inverted red triangles** indicate detection results that do not match the ground truth. Zoom in for more details.

method, YOLOv5. Fusion-Mamba and ICAFusion achieved the second best results after the proposed JTMDet in terms of the mAP50 (89.5% vs. 90.9%) and mAP (61.9% vs. 63.6%) scores, respectively.

4.5. Qualitative Analysis

To qualitatively demonstrate the performance of the proposed JTMDet, we visualized representative object detection results (shown in Figs. 5–7) on the FLIR, LLVIP and M³FD datasets and compared with other methods.

Comparisons on FLIR. As shown in Fig. 5, the proposed method outperformed all the other methods. Specifically, there are a number of missing detections in the detection results of CFT and ICAFusion, see for example, the pedestrians on the roadside and the cars occluded by the cyclists in Fig. 5 (a), and the pedestrians in the distance in Fig. 5 (b). However, the proposed method worked very well in these situations. Further, the FLIR dataset itself has complex lighting conditions and incomplete image registration. The fact that the proposed method can still work well under such conditions show that the method can effectively mine complementary information and enhance the object features under different lighting and low-quality image pair conditions.

Comparisons on LLVIP. From the results on the LLVIP dataset, we can see that the proposed method can handle occlusion or overlap problems very well. As shown in Fig. 6 (a), the pedestrian at the top of the image is severely occluded by the trees, and neither CFT nor ICAFusion was able

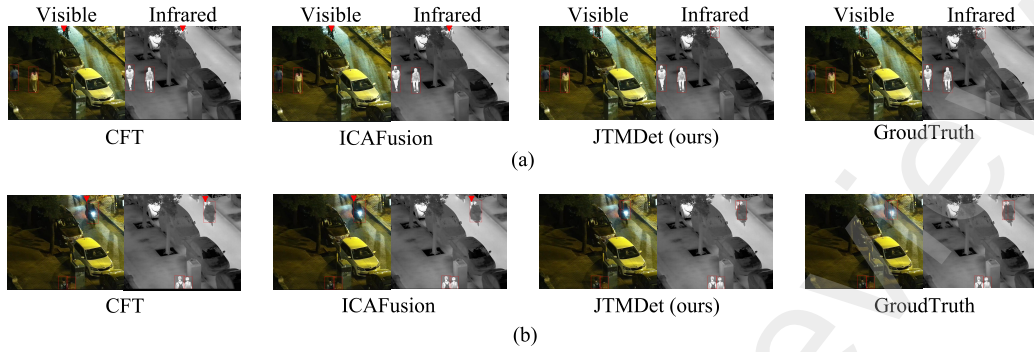


Figure 6: Qualitative analysis of CFT[1], ICAFusion[2] and the proposed method on LLVIP dataset. For simplicity, the categories of the predicted results are not shown, and different categories are marked with boxes of different colors. **Inverted red triangles** indicate detection results that do not match the ground truth. Zoom in for more details.

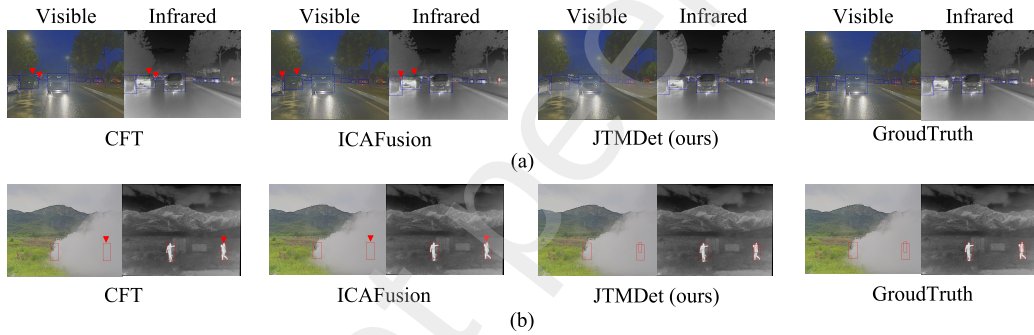


Figure 7: Qualitative analysis of CFT[1], ICAFusion[2] and the proposed method on M³FD dataset. For simplicity, the categories of the predicted results are not shown, and different categories are marked with boxes of different colors. **Inverted red triangles** indicate detection results that do not match the ground truth. Zoom in for more details.

520 to detect it. By contrast, the proposed method successfully detected the
 521 pedestrian. In Fig. 6 (b), the motorcyclists are overlapping one another, and
 522 cannot be distinguished by the visible light image alone, due to a dark and
 523 blurry background. Unlike CFT and ICAFusion, the proposed method can
 524 also work well in this case. These results show that the proposed method
 525 can effectively fuse information of the two modalities.

526 **Comparisons on M³FD.** As shown in Fig. 7 (a), the complex envi-
 527 ronment and insufficient lighting introduce a lot of interference information
 528 into the image, resulting in mis-detections for CFT and ICAFusion. In con-
 529 trast, the proposed method can detect all the objects, which shows that the

Table 4: Ablation studies of model components on the FLIR dataset

Modules			Metrics(%)		Param.(M)	Time(ms)
CMF	3DConv	CLF	mAP50	mAP		
			81.8	44.9	72.7	13.7
✓			83.2	45.9	165.4	26.5
	✓		81	45	72.7	14.2
		✓	82.9	45.6	144.6	17.7
✓	✓		84.1	46.2	165.4	26.6
✓		✓	83.6	46.1	237.3	29.2
	✓	✓	83.4	46.3	144.6	18.8
✓	✓	✓	84.7	46.5	237.4	30.9

proposed method eliminate the influence of interference through sufficient feature interaction. Similarly, in Fig. 7 (b), the proposed method was not hindered by the smoke in the visible light image, and successfully detected two overlapping targets that CFT and ICAFusion failed to detect.

4.6. Ablation Study

In this section, we use the FLIR dataset to ablate the the proposed JTMDet framework. First, we verify the effectiveness of the 3D convolutional fusion module, and then the CMF module and CLF modules in the staged fusion strategy. Next, we further explore the impact of the ratio of Mamba and Transformer layers, as well as the number of stacked JTM blocks. Finally, to verify that the effectiveness of the proposed JTMDet comes mainly from the proposed CMF and CLF modules and the JTM block, we replace the YOLOv5 model with a YOLOv8 model while keeping the other parts of the JTMDet framework unchanged.

Effectiveness of the CMF module, CLF module and 3D convolution module. We tested the effectiveness of these modules by alternately removing them from the architecture. Note that after removing the 3DConv module, we use element-wise addition instead. The results are summarized in Table 4. As we can seen from the table, after removing the CMF module, the performance of the architecture in terms of mAP50 and mAP dropped by 1.3% and 0.2%, respectively, which shows that features of different modalities can interact effectively through CMF module to generate better and more robust enhanced features. After removing the CLF module, the mAP50 and

Table 5: The impact of the ratio of Mamba and Transformer in JTM Blocks on the FLIR dataset

Ratio(M:T)	mAP50(%)	mAP(%)	Param.(M)	Time(ms)
3 : 0	84.1	46.5	231.2	31.8
2 : 1	84.7	46.5	237.4	30.9
1 : 2	84.3	46.1	243.4	27.5
0 : 3	84.2	46.3	249.5	24.8

mAP decreased by 0.6% and 0.3%, respectively. We believe that this is because CLF module enables different fine-grained features to enhance each other, thereby improving the quality of the obtained features. In addition, it also can be seen that the influence of CLF module is not as important as that of the CMF module. This is because the CLF module itself is designed to process the features enhanced by the CMF module. When the CMF module is removed, the quality of the features fed into the CLF module is compromised. After removing the 3D convolution fusion module, the mAP50 and mAP decreased by 1.1% and 0.4%, respectively. Interestingly, using 3D convolution alone does not bring a certain gain to the model (-0.8% mAP and +0.1% mAP), but it can bring additional benefits when used in conjunction with CMF and CLF modules. This shows that when the features of both modalities are of good quality, using the 3D convolution module instead of simple additive fusion can further improve the quality of the fused features. When the features of both modalities are not of good quality, directly using the 3D convolution module may introduce additional errors, making the quality of the fused features worse. At the same time, this result once again proves the key role of the CMF module in the entire model. Finally, as shown in the last row of Table 4, the combination of all these three modules leads the proposed JTMDet framework to its best performance.

The impact of the ratio of Mamba and Transformer layers in the JTM block. We studied the composition ratio of Mamba and Transformer layers in the JTM blocks. The results are shown in Table 5, it can be seen from the table that the architecture size increases and the running time gradually drops as more Transformer layers are used, but the performance of the model does not increase accordingly. The performance of the architecture reaches the optimum when the ratio of Mamba to Transformer layers is 2:1, achieving 84.7% and 46.5% in the mAP50 and mAP scores, respectively. This suggests that a balance between Mamba and Transformer is needed to fully

Table 6: The impact of the number of JTM blocks on the FLIR dataset

number	mAP50(%)	mAP(%)	Param.(M)	Time(ms)
1	83.7	46.3	156	23.5
2	84.7	46.5	237.4	30.9
3	83.7	46.1	318.4	36.4
4	83	45.5	399.9	44.3

Table 7: Generalization performance test of JTMDet with different networks on the FLIR dataset

Methods	Backbone	Data Type	mAP50(%)	mAP(%)
YOLOv5	YOLOv5	RGB	67.8	37.8
YOLOv5	YOLOv5	IR	73.9	39.5
Add-Fuison	YOLOv5	RGB+IR	81.8	44.9
JTMDet	YOLOv5	RGB+IR	84.7	46.5
YOLOv8	YOLOv8	RGB	66.3	28.2
YOLOv8	YOLOv8	IR	72.9	38.3
Add-Fusion	YOLOv8	RGB+IR	78.7	43.8
JTMDet	YOLOv8	RGB+IR	81.2	44.7

exploit their complementarity, So we finally form our JTMDet with the ratio of 2:1. In addition, we also observed that the order of the Transformer and Mamba layers in a JTM block is not important once their ratio is fixed.

The impact of the number of JTM blocks. As shown in Table 6, as the number of JTM blocks stacks increases, the number of model parameters and running time increase significantly. However, when the number of JTM blocks stacks exceeds two, the performance of the architecture decreases, suggesting that the architecture becomes overfitted. Therefore, we finally set the number of JTM blocks to two in this study.

Generalization performance test. In order to test the effect of integrating our proposed module into different base models, we further performed ablation tests on the structure of JTMDet on with YOLOv8 [65] replacing YOLOv5 [55]. The results are shown in Table 7. For the convenience of comparison, we also list the ablation experiment results on YOLOv5. Surprisingly, the performance of JTMDet with YOLOv8 is worse than that based on YOLOv5. This is because YOLOv8 limits the number of channels of its

feature maps in the backbone network in order to control the scale of the model, which makes its number of parameters far lower than JTMDet, and limits the modeling and feature extraction capabilities of the model¹. But it still can be seen from these experiments that for both YOLOv5 and YOLOv8, the proposed JTMDet framework performs consistently, suggesting that its performance is mainly attributed to the the proposed CMF and CLF modules.

5. Conclusion

In this paper, we propose a novel framework called the joint Transformer and Mamba detection (JTMDet). In order to make full use of image features of different modalities, we design a two-stage fusion strategy, including an intra-scale fusion stage and an inter-scale fusion stage. For the two-stage fusion strategy, we designed the CMF and CLF modules, both of which contain a hybrid structure of Transformer and Mamba, to reach an optimal trade-off between speed and accuracy performance. Extensive experiments on three public RGB-IR datasets show that the proposed JTMDet framework achieved state-of-the-art performance with higher inference efficiency. Our study confirms the potential of the hybrid structure of Transformer and Mamba in cross-modal feature fusion, and we hope that our work can contribute to the research of the hybrid structure of Transformer and Mamba.

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¹We used the YOLOv8 model v8.2.32 downloaded from [65]. However, please note that the YOLOv8 model we used in this ablation study may not be the same as the one used in Fusion-Mamba [48]. As the authors of Fusion-Mamba did not make their source code publicly available, it is hard for us to determine the exact version of the YOLOv8 model used in their study. But nonetheless, the experiments in Section 4.4 did show that the proposed method based on YOLOv5 is already competitive with Fusion-Mamba on both YOLOv5 and YOLOv8 versions.

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